

Literature-Implied Partial Answer to Problem 8.1 of arXiv:1406.5526

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Status. `literature_implied_answer` (partial subcase). This note records a direct consequence of the source paper's own theorems. It is not a full answer to the arbitrary non-reflexive case.

Source. A. Aviles, G. Plebanek, and J. Rodriguez, *Tukey classification of some ideals in omega and the lattices of weakly compact sets in Banach spaces*, arXiv:1406.5526v3 [1].

Original question. The first problem in the source paper's final "Open problems" section asks:

Is it true that $\mathcal{AK}(X) \sim \mathcal{AK}(B_X)$ for every non-reflexive X ?

In the PDF theorem numbering this is Problem 8.1.

Recorded partial answer. The following consequences are immediate from the comparison and classification results in [1].

- (a) If X is non-reflexive and $\mathcal{K}(B_X) \preceq \mathcal{AK}(B_X)$, then $\mathcal{AK}(X) \sim \mathcal{AK}(B_X)$.
- (b) If X is non-reflexive and SWCG, then $\mathcal{AK}(X) \sim \mathcal{AK}(B_X) \sim \omega$.
- (c) In ZFC, $\mathcal{AK}(X) \sim \mathcal{AK}(B_X)$ for every separable non-reflexive Banach space X containing no copy of ℓ^1 .
- (d) Under analytic determinacy, $\mathcal{AK}(X) \sim \mathcal{AK}(B_X)$ for every separable non-reflexive Banach space X .

Identification. For (a), Proposition 3.7(i) of [1], applied with $E = X$, gives $\mathcal{AK}(X) \preceq \mathcal{K}(X)$. Proposition 3.2(iii) gives $\mathcal{K}(X) \sim \mathcal{K}(B_X)$ whenever X is non-reflexive. Therefore

$$\mathcal{AK}(X) \preceq \mathcal{K}(X) \sim \mathcal{K}(B_X) \preceq \mathcal{AK}(B_X).$$

The reverse reduction $\mathcal{AK}(B_X) \preceq \mathcal{AK}(X)$ is Proposition 3.7(ii), so $\mathcal{AK}(X) \sim \mathcal{AK}(B_X)$.

For (b), Lemma 4.2(ii) of [1] gives a countable family of weakly compact sets in X which strongly generates every weakly compact subset of X . After replacing the family by increasing finite unions, the usual rank map $K \mapsto n$ such that $K \subset K_n + \varepsilon B_X$ is a Tukey reduction $\mathcal{AK}(X) \preceq \omega$. Theorem 4.3 of [1] gives $\mathcal{AK}(B_X) \sim \omega$ for non-reflexive SWCG spaces, and Proposition 3.7(ii) again gives the reverse lower bound.

For (c), Theorem 6.1 of [1] states that if X is separable and contains no copy of ℓ^1 , then $\mathcal{AK}(B_X) \sim \mathcal{K}(B_X)$, so (a) applies.

For (d), Theorem 8.1 of [1], under analytic determinacy, classifies $\mathcal{AK}(B_X)$ for separable X . In the non-reflexive case $\mathcal{AK}(B_X) \sim \omega$, (b) applies via Theorem 4.3. In the remaining non-reflexive cases ω^ω , $\mathcal{K}(\mathbb{Q})$, and $[\mathfrak{c}]^{<\omega}$, Theorem 8.1 itself states $\mathcal{AK}(B_X) \sim \mathcal{K}(B_X)$, so (a) applies.

Scope limitation. This does not settle Problem 8.1 for arbitrary non-reflexive Banach spaces. The standard scaling proof for $\mathcal{K}(X) \sim \mathcal{K}(B_X) \times \omega$ does not automatically adapt to $\mathcal{AK}$, because the Tukey tolerance δ must be fixed before the target radius bound is known.

References

- [1] A. Aviles, G. Plebanek, and J. Rodriguez, *Tukey classification of some ideals in omega and the lattices of weakly compact sets in Banach spaces*, arXiv:1406.5526v3.